1 The Group 17 elements are oxidising agents.

(a) (i) Explain how the Group 17 elements act as oxidising agents.

.....

..... [1]

(ii) Write an equation to show the reaction in which Cl_2 oxidises aluminium metal.

..... [1]

(d) Table 1.3 gives some information about $MgCl_2$ and $SiCl_4$.

Table 1.3

	$MgCl_2$	**$SiCl_4$**
electrical conductivity when liquid	conducts	does **not** conduct
observation when added to water	dissolves	vigorous reaction

(i) Explain the difference between the electrical conductivity of liquid $MgCl_2$ and of liquid $SiCl_4$. Refer to bonding and relevant particles in your answer.

.....

.....

.....

.....

..... [2]

(ii) Suggest the pH of the solutions that form when each chloride is added to water.

$MgCl_2$ ..... $SiCl_4$ .....
[2]

2 Aluminium oxide, Al_2O_3, and phosphorus(V) oxide, P_4O_{10}, are both used as reagents and catalysts.

(a) The melting point of Al_2O_3 is 2072 °C. The melting point of P_4O_{10} is 340 °C.

Explain the difference in the melting points of these two compounds.

.....

.....

.....

..... [3]

(iii) Write an equation to show the reaction of P_4O_{10} with an excess of water.

..... [1]

1 Chromium, Cr, and its compounds are widely used in many chemical reactions.

(a) Cr exists as four stable isotopes.

(i) The most common isotope of Cr is chromium-52.

Determine the number of protons, neutrons and electrons in an atom of chromium-52.

number of protons ..... neutrons ..... electrons .....
[1]

(iii) $CrO_4^{2-}(aq)$ ions are yellow and $Cr_2O_7^{2-}(aq)$ ions are orange.

State what is observed when the following changes are made to an equilibrium mixture of acidified $CrO_4^{2-}(aq)$ and $Cr_2O_7^{2-}(aq)$ ions.

Explain your answers.

- The equilibrium mixture is warmed gently.

  observation .....

  explanation .....

  .....

  .....

- Dilute HCl(aq) is added to the equilibrium mixture.

  observation .....

  explanation .....

  .....

  .....
[4]

(e) Chromium(IV) fluoride, CrF_4, is a covalent molecule that shows similar chemical properties to SiCl_4.

Suggest the type of reaction that occurs when CrF_4 is placed in water.

Construct a relevant equation for this reaction.

type of reaction .....

equation .....
[2]

2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) Fig. 2.1 shows the variation in atomic and ionic radii of the Period 3 elements Na to C*l*.

The ionic radius of Si is **not** shown.

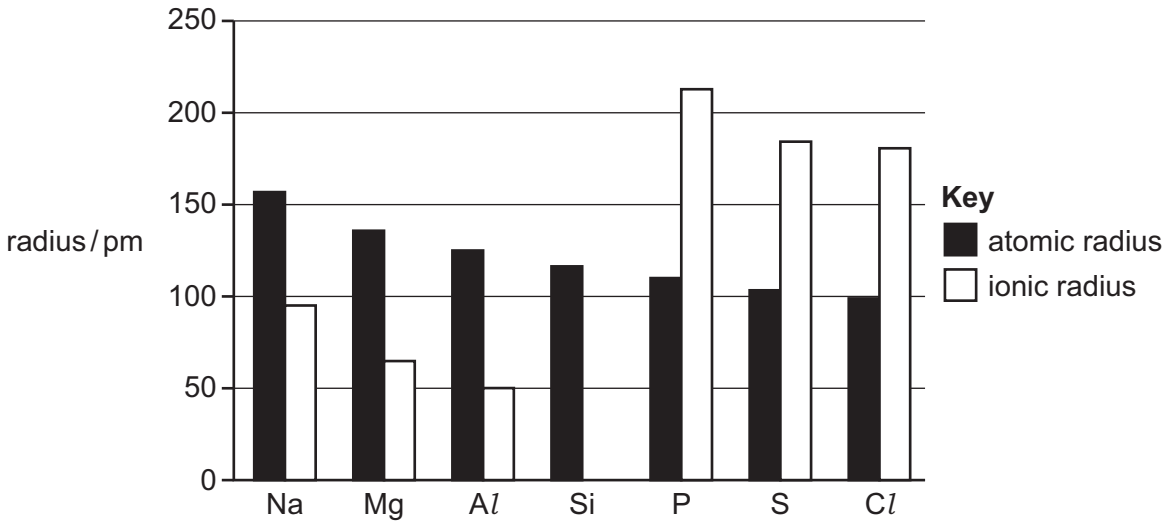


Fig. 2.1

(i) Explain the trend shown in the atomic radii of the Period 3 elements Na to C*l*.

.....

.....

.....

..... [2]

(ii) Explain why there is a large difference in the ionic radii of A*l* and P.

.....

.....

.....

..... [2]

(b) Table 2.1 gives some information about some of the Period 3 oxides.

Row B gives the pH of the solution that forms when the Period 3 oxide is added to water.

Table 2.1

	formula of Period 3 oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_3
A	oxidation number of Period 3 element			+3			
B	pH of solution			—	—		

(i) Complete Table 2.1. [2]

(ii) State why there is no data given in row B for Al_2O_3 and SiO_2.

..... [1]

(c) **(i)** Write an equation for the reaction of Na_2O with dilute hydrochloric acid.

..... [1]

(ii) Construct an equation for the reaction of Al_2O_3 with a base to form $NaAlO_2$.

..... [1]

2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) **(i)** Explain why the elements Na to A*l* are good electrical conductors.

.....

..... [1]

(ii) Explain why the elements P, S and C*l* do not conduct electricity.

.....

..... [1]

(b) Fig. 2.1 shows the variation in melting point of the Period 3 elements Si to C*l*.

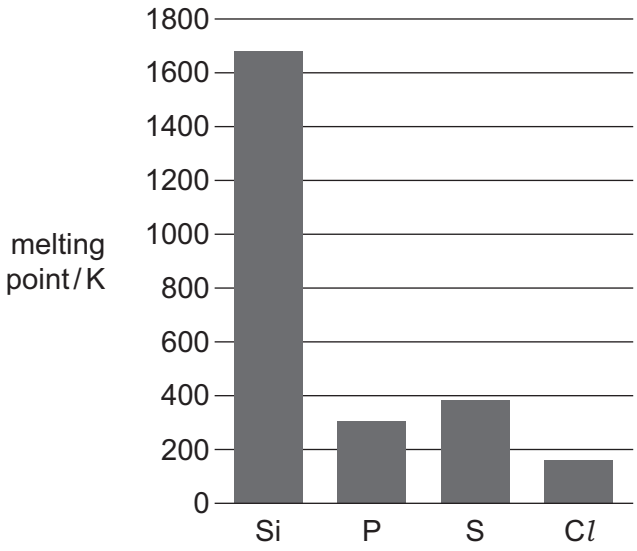


Fig. 2.1

The Period 3 elements Si to C*l* are all non-metals.

Explain why there is a large difference between the melting point of Si and the melting points of P, S and C*l*.

.....

.....

.....

..... [2]

(c) Table 2.1 gives some information about some Period 3 chlorides.

Row B refers to the pH of the solution that forms when the Period 3 chloride is added to water.

Table 2.1

	formula of Period 3 chloride	NaCl	$MgCl_2$	$AlCl_3$	$SiCl_4$	PCl_5
A	oxidation number of element bonded to Cl					
B	pH of solution		6.5			
C	bonding			ionic		
D	structure			giant		

Complete Table 2.1.

You may use the following abbreviations.

I = ionic, C = covalent, M = metallic
G = giant, S = simple

[4]

(d) **(i)** Write an equation for the formation of $AlCl_3$ from its elements.

.....

..... [1]

(ii) Write an equation for the formation of H_3PO_4 from PCl_5.

..... [1]

1 Chromium, Cr, and its compounds are widely used in many chemical reactions.

(a) Cr exists as four stable isotopes.

(i) The most common isotope of Cr is chromium-52.

Determine the number of protons, neutrons and electrons in an atom of chromium-52.

number of protons ..... neutrons ..... electrons .....
[1]

(iii) $CrO_4^{2-}(aq)$ ions are yellow and $Cr_2O_7^{2-}(aq)$ ions are orange.

State what is observed when the following changes are made to an equilibrium mixture of acidified $CrO_4^{2-}(aq)$ and $Cr_2O_7^{2-}(aq)$ ions.

Explain your answers.

- The equilibrium mixture is warmed gently.

  observation .....

  explanation .....

  .....

  .....

- Dilute HCl(aq) is added to the equilibrium mixture.

  observation .....

  explanation .....

  .....

  .....
[4]

(e) Chromium(IV) fluoride, CrF_4, is a covalent molecule that shows similar chemical properties to SiCl_4.

Suggest the type of reaction that occurs when CrF_4 is placed in water.

Construct a relevant equation for this reaction.

type of reaction .....

equation .....
[2]

2 The Period 3 elements show trends in physical and chemical properties across the period.

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The ionic radius of Si is **not** shown.

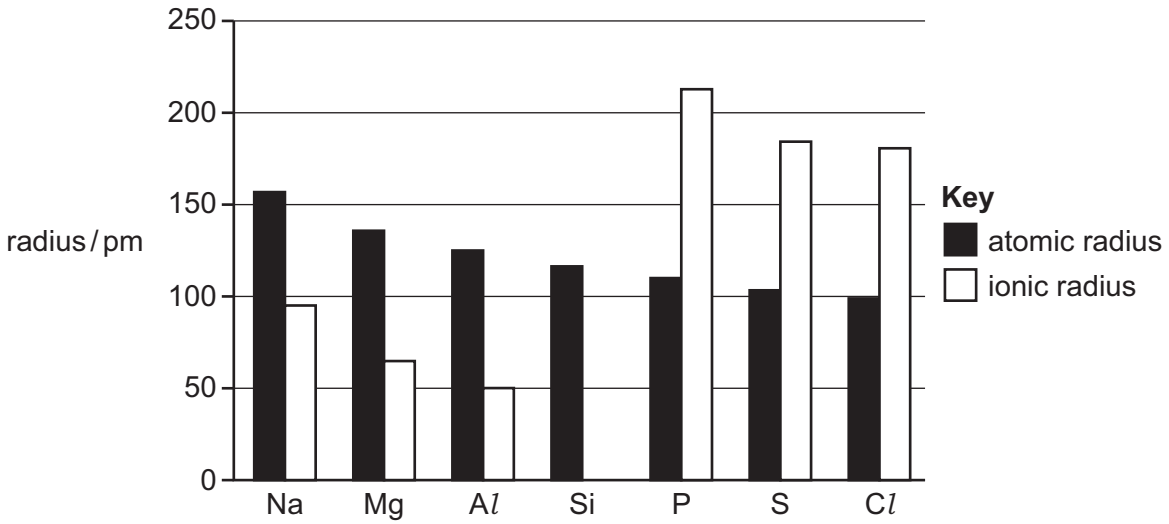


Fig. 2.1

(i) Explain the trend shown in the atomic radii of the Period 3 elements Na to C*l*.

.....

.....

.....

..... [2]

(ii) Explain why there is a large difference in the ionic radii of A*l* and P.

.....

.....

.....

..... [2]

(b) Table 2.1 gives some information about some of the Period 3 oxides.

Row B gives the pH of the solution that forms when the Period 3 oxide is added to water.

Table 2.1

	formula of Period 3 oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_3
A	oxidation number of Period 3 element			+3			
B	pH of solution			—	—		

(i) Complete Table 2.1. [2]

(ii) State why there is no data given in row B for Al_2O_3 and SiO_2.

..... [1]

(c) **(i)** Write an equation for the reaction of Na_2O with dilute hydrochloric acid.

..... [1]

(ii) Construct an equation for the reaction of Al_2O_3 with a base to form $NaAlO_2$.

..... [1]

1 Elements in Period 3 of the Periodic Table show trends in their properties.

(a) Complete Table 1.1 by identifying the lattice structures of the crystalline solids of Mg, Si and P.

Table 1.1

element	Mg	Si	P
lattice structure in crystalline solid			

[3]

(b) The relative electrical conductivities of the Period 3 elements are shown in Fig. 1.1.

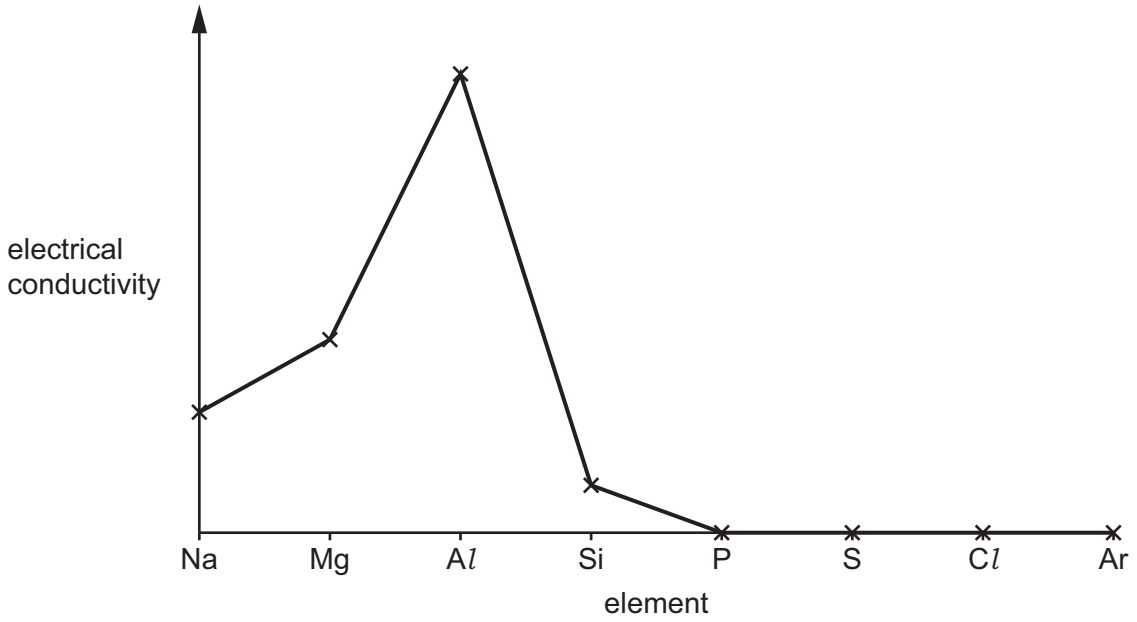


Fig. 1.1

Explain why there is an increase in conductivity from Na to Al and why P, S and Cl are non-conductors of electricity.

.....

.....

.....

.....

..... [2]

(d) Most Period 3 elements react with oxygen to form oxides.

(i) Write an equation for the reaction of phosphorus with oxygen.

..... [1]

(ii) Write an equation for the reaction of aluminium oxide with an excess of aqueous sodium hydroxide.

..... [1]

2 **(a)** Fig. 2.1 shows the covalent bonds and lone pairs of electrons in a molecule of SCl_2.

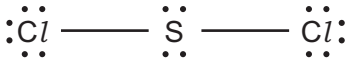


Fig. 2.1

(i) Use VSEPR theory to predict the shape of a molecule of SCl_2.

..... [1]

(b) SCl_2 is a red liquid that reacts with water.

$$2SCl_2(l) + 3H_2O(l) \rightarrow H_2SO_3(aq) + S(s) + 4HCl(aq)$$

(i) Describe **two** observations, other than temperature change, that are made when an excess of water is added to SCl_2.

.....

..... [2]

(ii) When SCl_2 reacts with water, the SCl_2 is broken down and a disproportionation reaction occurs.

State the oxidation numbers of sulfur in SCl_2 and H_2SO_3.

oxidation number of S in SCl_2 .....

oxidation number of S in H_2SO_3 .....
[2]

(iii) State the general name to describe the type of reaction that occurs when a substance is broken down by water.

..... [1]

2 The chemical properties of oxides are related to the chemical bonding present in these compounds.

(a) A Period 3 oxide produces a solution with a pH greater than 10 when it is added to water.
State the formula of the oxide.

..... [1]

(b) P_4O_{10} is added to an excess of aqueous NaOH.
Write an equation to describe the reaction.

..... [1]

(c) Table 2.1 shows the melting points of some oxides.

Table 2.1

oxide	melting point / °C
SO_2	–73
H_2O	0
SO_3	17
SiO_2	1610
Na_2O	1132
MgO	2852
Al_2O_3	2072

(i) Identify the oxide from Table 2.1 that contains the element with the highest oxidation number.

..... [1]

(e) Al_2O_3 is a white amphoteric compound.

(i) State the formula of the aluminium-containing species produced when Al_2O_3 reacts with NaOH(aq).

..... [1]

(ii) State the formula of the aluminium-containing salt produced when Al_2O_3 reacts with H_2SO_4(aq).

..... [1]

4 The reaction between $Na_2S_2O_3(aq)$ and $HCl(aq)$ is monitored at constant temperature.

$$Na_2S_2O_3 + 2HCl \rightarrow 2NaCl + SO_2 + S + H_2O$$

Fig. 4.1 shows how the concentration of $HCl(aq)$ varies with time.

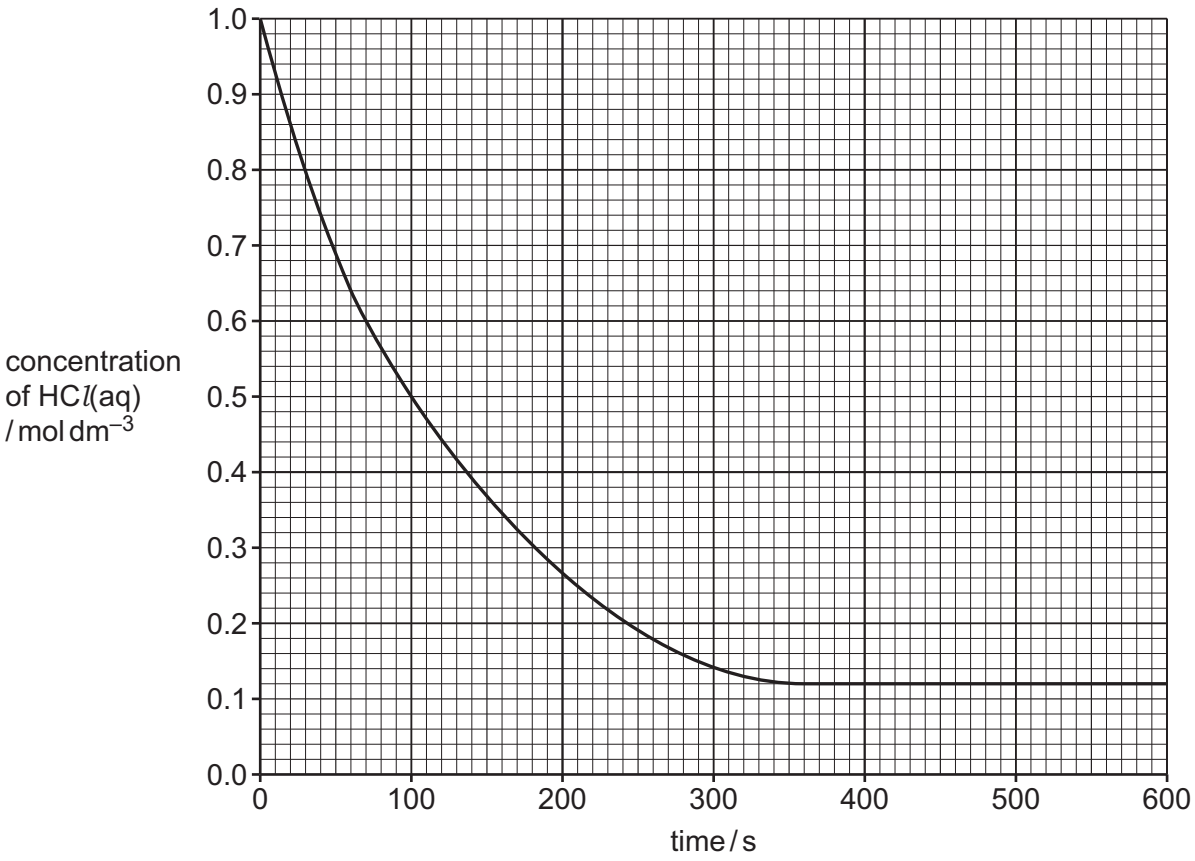


Fig. 4.1

(a) (i) Use Fig. 4.1 to find the average rate of change of concentration of $HCl(aq)$ in this reaction between 0–100 seconds and between 400–500 seconds. Include units in your answers.

0–100 seconds ..... units .....

400–500 seconds ..... units .....

[2]

(v) Explain why the rate of reaction **cannot** be monitored accurately by measuring the volume of $SO_2(g)$ produced in this reaction.

.....

..... [1]

1 **(a)** Diamond and graphite are both crystalline solids made from carbon atoms. Graphite conducts electricity. Diamond does **not** conduct electricity.

(i) Name the type of lattice structure present in the crystalline solids diamond and graphite.

..... [1]

(iii) Explain why diamond does **not** conduct electricity.

.....

..... [1]

(b) Separate samples of phosphorus(V) chloride and silicon(IV) chloride are each added to an excess of cold water.

(i) Write an equation for each reaction.

.....

..... [2]

(ii) Describe the appearance of phosphorus(V) chloride and silicon(IV) chloride at room temperature.

phosphorus(V) chloride .....

silicon(IV) chloride .....
[2]

(iii) Compare the appearance of the mixtures produced when each reaction is complete.

.....

..... [2]

(c) Some oxides are amphoteric.

(i) Describe what is meant by amphoteric.

.....

..... [1]

(ii) Identify the formula of a Period 3 oxide that is amphoteric.

..... [1]

1 **(a)** Solid sodium conducts electricity. Sodium oxide conducts electricity when molten but not when solid.

(i) Name the type of bonding present in sodium and in sodium oxide.

bonding in sodium .....

bonding in sodium oxide .....
[1]

(iii) Explain why sodium oxide conducts electricity when molten.

.....

..... [1]

(b) Separate samples of sodium and sodium oxide are each added to an excess of cold water.

(i) Write an equation for the reaction of sodium with cold water.

..... [1]

(ii) Write an equation for the reaction of sodium oxide with cold water.

..... [1]

(iii) Complete Table 1.1.
Do **not** refer to temperature changes when considering observations for these reactions.

Table 1.1

	sodium	sodium oxide
one similarity in observation on addition to cold water		
one difference in observation on addition to cold water		

[2]

(c) State the oxidation number of the Period 3 elements bonded to Cl in NaCl and PCl_5.
Explain the difference in the oxidation number.

.....

.....

..... [2]

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

(a) Chlorine forms three different compounds with phosphorus.

The most common compounds are PCl_3 and PCl_5.

(i) Complete Table 1.1.

Table 1.1

compound	oxidation number of P	oxidation number of Cl
PCl_3		
PCl_5		

[2]

(iii) The third compound of phosphorus and chlorine, **W**, has a relative molecular mass, M_r, between that of PCl_3 and PCl_5. The compound contains 69.6% by mass of chlorine.

Determine the molecular formula of **W**.

molecular formula of **W** = ... [2]

(iv) **W** is a liquid at room temperature and pressure. It reacts vigorously with water to form an acidic solution.

Suggest the structure and bonding in **W**. Explain your answer.

.....

.....

..... [2]

(b) Fig. 1.1 shows a reaction scheme involving P_4 and Cl_2.

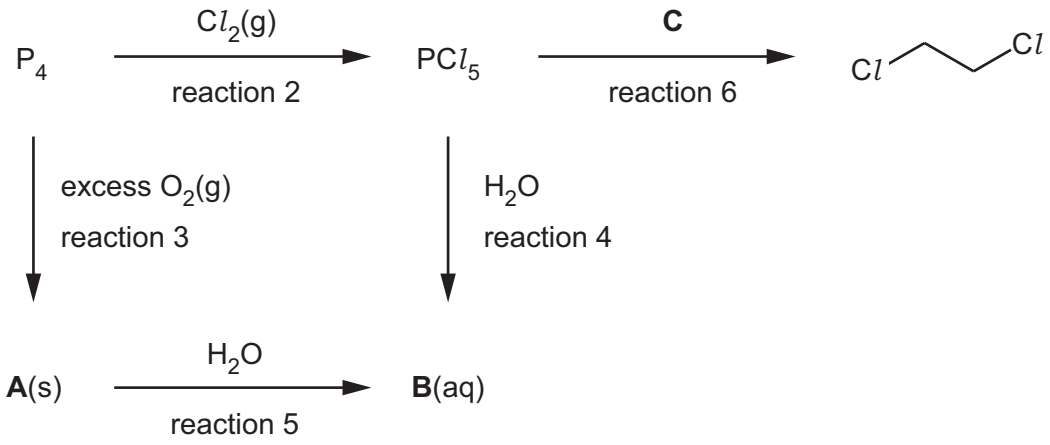


Fig. 1.1

(i) Suggest what is observed in reaction 2.

..... [1]

(iii) Write an equation for the formation of **A** in reaction 3.

..... [1]

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

(i) Complete Table 3.1. [3]

(e) Silicon readily reacts with elements of high electronegativity.

(i) Write an equation for the formation of $SiCl_4$ from its constituent elements.

..... [1]

(ii) Describe what is observed when a small sample of $SiCl_4$ is added to water.

.....

..... [1]

(f) Tin forms an amphoteric oxide, SnO_2.

Suggest the formula of the tin compound that forms when SnO_2 reacts with H_2SO_4 in an acid–base reaction.

..... [1]

2 Oxygen is a Group 16 element.

(a) (i) Write equations for the following reactions.

- sodium and oxygen

.....

- sulfur and oxygen

.....

[2]

(iii) The maximum oxidation state of the Period 3 elements in their oxides varies across the period.

State and explain the variation.

.....

.....

.....

[2]

(b) H_2O reacts with both inorganic and organic compounds.

(i) Complete Table 2.1 to give details of the reactions of some Period 3 oxides with H_2O.

Table 2.1

Period 3 oxide	product of reaction with H_2O	pH of solution formed
	$Mg(OH)_2$	
P_4O_{10}		

[2]

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

(i) Complete Table 3.1. [3]

(e) Silicon readily reacts with elements of high electronegativity.

(i) Write an equation for the formation of $SiCl_4$ from its constituent elements.

..... [1]

(ii) Describe what is observed when a small sample of $SiCl_4$ is added to water.

.....

..... [1]

(f) Tin forms an amphoteric oxide, SnO_2.

Suggest the formula of the tin compound that forms when SnO_2 reacts with H_2SO_4 in an acid–base reaction.

..... [1]

1 **(a)** **(i)** Explain the lack of reactivity of nitrogen gas, $N_2(g)$.

.....

.....

..... [2]

(b) **(i)** A sample of Al reacts with an excess of Cl_2.

State the oxidation number of Al in the product of the reaction.

oxidation number of Al ..... [1]

(ii) State what determines the maximum oxidation number of the Period 3 elements in their oxides.

.....

..... [1]

(c) Separate samples of aluminium oxide, Al_2O_3, and phosphorus(V) oxide, P_4O_{10}, react with an excess of NaOH(aq) at room temperature.

(i) Give the state of Al_2O_3 and P_4O_{10} at room temperature.

Al_2O_3 .....

P_4O_{10} .....
[1]

(ii) Write an equation for the reaction of each oxide with an excess of NaOH(aq) at room temperature.

Al_2O_3 + ...

P_4O_{10} + ...
[2]

1 **(a)** Complete Table 1.1 using relevant information from the Periodic Table.

Table 1.1

	nucleon number	proton number	number of electrons
Mg^{2+}	24		
Al^{3+}	27		

[2]

(d) Fig. 1.1 shows the variation in melting point of some Period 3 elements in their standard states at room temperature and pressure.

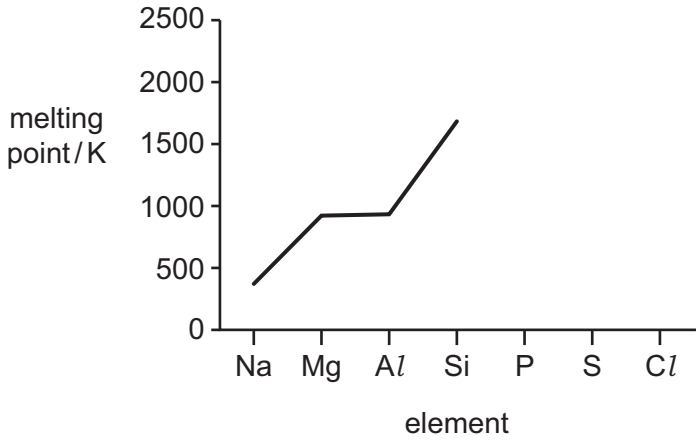


Fig. 1.1

(i) Explain why Si has a high melting point.

.....

.....

..... [1]

(ii) Complete Fig. 1.1 to show the variation in the melting points of the elements P, S and Cl.
[2]

(e) Two Period 3 elements react with an excess of oxygen at room pressure.

(i) Complete Table 1.2.

Table 1.2

1	2	3
Period 3 element	state of oxide at room temperature and pressure	approximate pH of solution made when oxide is added to water
Na		
S		

[2]

(ii) The solutions made in column 3 of Table 1.2 are mixed together.
Name the type of reaction that occurs.

..... [1]

(iii) Write an equation to describe the reaction between P_4O_{10} and an excess of water.

..... [1]

(f) Aluminium hydroxide is amphoteric.

(i) Explain what is meant by amphoteric.

.....

..... [1]

(ii) Write an equation to describe the reaction that occurs when aluminium hydroxide, $Al(OH)_3$, reacts with NaOH(aq).

..... [1]

2 **(a)** Sulfur chloride, SCl_2, is a liquid at room temperature. When SCl_2 is added to water, misty fumes are seen and a solution is made that turns universal indicator red.

(i) Identify the type of reaction that occurs when SCl_2 is added to water.

..... [1]

(ii) Name a chloride of a different Period 3 element that is also a liquid at room temperature and produces misty fumes when added to water.

..... [1]

2 Chlorine, Cl_2, reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

(a) Complete Table 2.1. [3]

(c) An excess of Cl_2 reacts with phosphorus to form PCl_5.

(i) PCl_5 is a simple molecule in the gas phase.

It also exists in a solid form as two ions, PCl_4^+ and PCl_6^-.

Complete Table 2.2 to identify the shapes of each of these species.

Table 2.2

species	PCl_5	PCl_4^+	PCl_6^-
shape		tetrahedral	

[2]

(ii) PCl_5 reacts with **J** to form H_3PO_4.

Identify **J** and state the type of reaction.

J ..... type of reaction .....

[2]

1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to C*l*.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	C*l*
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol^{-1}	1060		1260
formula of most common anion	P^{3-}	S^{2-}	Cl^{-}

(a) **(i)** Complete Table 1.1 to show the number of electrons in the 3p subshell and the total number of unpaired electrons in an atom of P, S and C*l*. [2]

(d) $POCl_3$ shows similar chemical properties to PCl_5.

$POCl_3$ has a melting point of 1°C and a boiling point of 106 °C.

$POCl_3$ reacts vigorously with water, forming misty fumes and an acidic solution.

(i) Explain how the information in **(d)** suggests the structure and bonding of $POCl_3$ is simple covalent.

.....

.....

..... [2]

(ii) Construct an equation for the reaction of $POCl_3$ with water.

$POCl_3$ + ... → .....
[1]

1 The elements silicon, phosphorus and sulfur are in Period 3 of the Periodic Table.

(a) (i) Describe the variation in atomic radius from silicon to sulfur.

..... [1]

(ii) The melting point of silicon is 1410 °C. The melting point of sulfur is 113 °C.

Explain this difference.

.....

.....

.....

..... [3]

(b) Table 1.1 shows some properties of the elements Si to S.

The first ionisation energy of P is **not** shown.

Table 1.1

property	Si	P	S
total number of electrons in s subshells			
total number of electrons in p subshells			
first ionisation energy / kJ mol^{-1}	786		1000
formula of most common chloride	$SiCl_4$	PCl_5	SCl_2

(i) Complete Table 1.1 to show the total number of s and p electrons in an atom of Si, P and S.

[2]

(iv) $SiCl_4$ and PCl_5 each react with water, forming misty fumes.

Identify the chemical responsible for the misty fumes.

..... [1]

3 Phosphoric(V) acid, H_3PO_4, is used in both inorganic and organic reactions.

(a) H_3PO_4 is made in a two-step process from phosphorus.

step 1 Phosphorus reacts with an excess of oxygen to form a white solid.

step 2 The white solid then reacts with water to form H_3PO_4.

(i) Write an equation for each step.

step 1 .....

step 2 .....

[2]

1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to C*l*.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	C*l*
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol^{-1}	1060		1260
formula of most common anion	P^{3-}	S^{2-}	Cl^{-}

(a) **(i)** Complete Table 1.1 to show the number of electrons in the 3p subshell and the total number of unpaired electrons in an atom of P, S and C*l*. [2]

(d) $POCl_3$ shows similar chemical properties to PCl_5.

$POCl_3$ has a melting point of 1 °C and a boiling point of 106 °C.

$POCl_3$ reacts vigorously with water, forming misty fumes and an acidic solution.

(i) Explain how the information in **(d)** suggests the structure and bonding of $POCl_3$ is simple covalent.

.....

.....

..... [2]

(ii) Construct an equation for the reaction of $POCl_3$ with water.

$POCl_3$ + ... → .....
[1]

2 **(a)** The reaction of pure aluminium is only observed if the aluminium oxide layer is removed first. When pure aluminium is added to cold water, bubbles of gas are seen.

(i) State **one** property of aluminium oxide that explains why an aluminium object does **not** react with cold water until the aluminium oxide layer is removed.

..... [1]

(ii) Write an equation, with state symbols, for the reaction of aluminium oxide with an excess of NaOH(aq).

..... [2]

(iii) Name **one** other Period 3 element that also produces bubbles of gas when added to cold water.

..... [1]

(c) Aluminium chloride and phosphorus chloride are both white solids.

(i) State the maximum oxidation number of aluminium and of phosphorus in these solid chloride salts.

maximum oxidation number of aluminium .....

maximum oxidation number of phosphorus .....

[1]

(ii) State why the maximum oxidation number of aluminium is different from that of phosphorus.

.....

..... [1]

(iii) Write an equation for the reaction of solid phosphorus chloride and excess water.

..... [1]

(iv) Name the type of reaction that occurs when aluminium chloride is added to water.

..... [1]

(v) Explain why the solution produced after aluminium chloride is added to water has a pH of 1–2.

.....

..... [1]

1 The melting points of some solids are shown in Table 1.1.

Table 1.1

solid	melting point / K
magnesium	923
phosphorus	317
sodium chloride	1074
sulfur	392

(a) (i) State the type of bonding present in magnesium and in sodium chloride.

bonding in magnesium .....

bonding in sodium chloride .....
[1]

(iii) Explain the difference in the melting points of phosphorus and sulfur in terms of structure and bonding.

.....

.....

..... [2]

2 Chlorine is a reactive element. It forms many compounds.

(a) **(i)** Complete Table 2.1 to show the maximum oxidation number of the elements Na to P in their chlorides.

Table 2.1

element	Na	Mg	A*l*	Si	P
maximum oxidation number					

[1]

(ii) State what determines the maximum oxidation number of elements in Period 3.

.....

..... [1]

(b) An excess of cold water is added to the chloride of silicon.

(i) Write an equation for the reaction between an excess of cold water and the chloride of silicon.

..... [1]

(ii) Suggest the pH of the solution produced in **(b)(i)**.

..... [1]

(c) An excess of cold water is added to the chloride of phosphorus.

(i) Write an equation for the reaction between an excess of cold water and the chloride of phosphorus.

..... [1]

(ii) Suggest the pH of the solution produced in **(c)(i)**.

..... [1]

1 Tellurium is an element in Group 16. The most common isotope of tellurium is ^{130}Te. Its electronic configuration is [Kr] $4d^{10}\ 5s^2\ 5p^4$.

(a) Complete Table 1.1.

Table 1.1

	nucleon number	number of neutrons	number of electrons
^{130}Te			

[3]

(f) TeF_x reacts with water to form tellurium hydroxide and HF. The oxidation number of tellurium does **not** change during this reaction.

(i) Construct an equation for the reaction of TeF_x with water.

..... [1]

(ii) Name the type of reaction that occurs when TeF_x reacts with water.

..... [1]

3 Chlorine is a very reactive element.

(a) Chlorine reacts with silicon to form silicon(IV) chloride.
Describe the appearance of silicon(IV) chloride at room temperature and pressure. State its structure and bonding.

appearance .....

structure and bonding .....
[2]

(b) Samples of magnesium chloride and phosphorus(V) chloride are added to separate beakers of cold water.

Complete Table 3.1. Ignore temperature changes when considering observations for these reactions.

Table 3.1

	magnesium chloride	phosphorus(V) chloride
appearance at room temperature		
one similarity in observation on addition to cold water		
one difference in observation on addition to cold water		
pH of final solution		

[4]

1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy / kJ mol^{-1}	1310	519	1090	1310	1000
second ionisation energy / kJ mol^{-1}	—	7300	2350	3390	2260

(a) (i) Define electronegativity.

.....

..... [1]

(d) CO_2 and SO_2 are acidic gases.

(i) Write an equation for the reaction of SO_2 with H_2O.

..... [1]

(ii) Write an equation for the reaction of SO_2 with NaOH.

..... [1]

(iii) Construct an equation for the reaction of CO_2 with $Mg(OH)_2$.

..... [1]

2 The Group 2 elements Mg to Ba are all silvery-white reactive metals.

(a) (i) Draw a labelled diagram to show the bonding and structure of the Group 2 metals at room temperature.

[2]

(ii) Explain why Mg has a higher electrical conductivity than Na.

.....

..... [1]